

Subject: [EXTERNAL] Decision on manuscript NCOMMS-25-61709-T
Date: Monday, November 24, 2025 at 1:09:48 AM Eastern Standard Time
From: tingheithomas.chan@springernature.com
To: Thaker, Vidhu V.

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Dear Professor Thaker,

Your manuscript entitled "Baseline metabolomic profile as potential biomarker for weight change after Roux-en-Y gastric bypass (RYGB) surgery" has now been seen by 4 referees, whose comments are appended below. Considering their advice I regret to inform you that we cannot publish your manuscript in Nature Communications.

You will see that, while the reviewers find your work of interest, they raise substantive concerns that cast doubt on the advance your findings represent over earlier work and the strength of the novel conclusions that can be drawn at this stage. Unfortunately, these reservations are sufficiently important to preclude publication of this study in Nature Communications.

Although we cannot publish your paper, it may be appropriate for another journal in the Nature Portfolio. If you wish to explore the journals and transfer your manuscript please use our [manuscript transfer portal](#). You will not have to re-supply manuscript metadata and files, unless you wish to make modifications. For more information, please see our [manuscript transfer FAQ](#) page.

I am sorry that we cannot be more positive on this occasion and thank you for the opportunity to consider your work.

Best regards,

Thomas Chan
Editor
Nature Communications

Reviewers' comments:

Reviewer #1 (Remarks to the Author):

Gu et al. present data from an untargeted metabolomics screen and a subsequent metabolite risk score that predicts weight regain in obese individuals after Roux-en-Y gastric bypass surgery. Weight regain after weight loss is observed in all weight loss methods (diet based, GLP1 agonis treatment and surgery) Because of weight regain many obese subjects enter into a spiral of repeated weight loss / weight regain cycles (yo-yoing). This is major concern for patients undergoing these treatments, causing stress and anxiety. From a clinical point of view, weight

regain cancels many of the metabolic improvements gained through the weight loss. As outlined in the article the propensity to regain or maintain weight loss is highly variable. Environmental factors certainly play a role but even correcting for those some variability persists. Hence, understanding the physiological factors that underly the susceptibility to regain weight might open new treatment avenues for better long term weight maintenance after weight loss. In this respect the article addresses an important, unsolved issue.

Specific remarks:

Selection of metabolite features for the risk score: Out of 541 QCed metabolites the authors select 62 features associated with “weight regain” using a selection of 80 weight regainers vs 80 weight stable subjects. I have several questions concerning this selection:

- No correction for multiple testing was applied in the selection of the 62 features (apparently the authors did use FDR to correct but only five of the 62 features were significantly associated with a $FRD < 5\%$). Even if we don't consider the five metabolites with a $FRD < 5\%$, this would still mean an expected 26 false positive findings (for 536 tests) from the initial association out of the 62 features. I would like to see the overall p-value distribution for the 541 metabolites.
- It would be helpful to have the correlation matrix for all of the 62 selected features as a supplementary figure/table.
- The selection of extreme groups might cause a power problem. The authors should consider to use the whole cohort and treat weight regain as a quantitative trait (QT) instead of the nested case-control design.
- In the QT design the analysis could also be corrected for initial weight loss (the weight regainers in the nested study lost less weight) to identify features for weight regain independent of initial weight loss.
- For either study design there should be a consistent approach to correct for multiple testing and only features meeting the multiple testing threshold should be included in the risk score.

Replication studies for the metabolite risk score:

- The replication data set is very small with only 21 subjects in each group. To be really confident in the replication one would need a data set that would at least match the initial discovery cohort sample size.

The Estonian biobank study is interesting as it might indicate that the markers might generally be associated with body weight. Again this could be tested using the whole sample set and performing a QT analysis with the metabolic score. One caveat of the experiment is that the markers used are not exactly the same as for the initial score. This raises some questions about the specificity of these markers for a score to assess weight regain.

GWAS study: The authors report a mQTL GWAS study for the metabolites included in the risk score. They identify a number of SNPs associated with metabolite levels and speculate if this could contribute to the differences seen in the score with weight regain.

- Although the authors report two apparently novel associations the GWAS study is generally too small to really provide an idea about the genetic architecture underlying the metabolite variability. The authors should discuss the relative relevance of their findings in the context of the much bigger mQTL GWAS studies (see for example Hysi et al. doi: 10.3390/metabo12010061 or Karjalainen et al. doi.org/10.1038/s41586-024-07148-y).
- To appreciate the power from the modest sample it would be interesting to know the p-values of SNPs that have been reported for kynurenine levels and that did not reach genome wide

significance in the Framingham sample.

- For the GWAS data is shown in extended figure 8-c, please show the effect sizes (beta values). The row with the chromosomal position could be deleted.
- The real value of the genetic data in the context of the main scientific question of the paper i.e. the metabolite levels are associated with and possibly drive the susceptibility to regain weight is the possibility to test for potential causality by performing a Mendelian Randomization study. The authors acknowledge this fact in their discussion but state that the MR is “out of scope”. I would argue that it is a key question to be answered. It would strengthen the proposed conclusions considerably.

Reviewer #2 (Remarks to the Author):

The manuscript of Gu and colleagues investigated whether baseline untargeted metabolomics could identify metabolites that predict long-term weight outcomes following bariatric surgery, using a nested case-control design. They further examined whether the prioritized metabolites were associated with obesity-related phenotypes in independent cohorts and conducted GWAS analyses to explore whether some metabolites linked to long-term weight outcomes have a genetic basis. The authors identified 62 metabolites nominally associated with %EBW5 and constructed a metabolic risk score (MetRS) based on a 13-metabolite panel. This score was evaluated in relation to long-term weight outcomes, particularly as a predictor of weight regain seven years post-surgery.

Overall, the authors provide a thorough description of their methods and results and make a commendable effort to validate the MetRS across different cohorts. However, as the metabolic profiling was conducted only at baseline, it remains difficult to determine whether surgery subsequently reduced or increased these metabolites, limiting conclusions about the persistence or evolution of the metabolic signature post-surgery.

Major concerns

- 1) In the absence of metabolite measurements at follow-up (i.e., after RYGB), it is not clear how the effect on metabolites is assessed. Could the authors clarify whether the predictive value of the baseline profile for weight gain carries meaningful implications in this context? It might also be useful to discuss what is expected to happen to the metabolite components of MetRS 5/7 years post-surgery
- 2) EBW is operationalized as weight above an “ideal” BMI of 25 kg/m². This assumption may be overly rigid and not reflective of interindividual differences in body composition, ethnicity, or health risk. Reliance on this cutoff could bias class allocation and limit generalizability.
- 3) The overwhelming dominance of Class 3 (86.9%) raises concerns about the generalisability of any outcomes from this study. It appears they will only apply to individual with EBW up around the 100kg mark and that these results do not apply to the vast majority of patients.
- 4) By focusing only on Classes 1 and 2, which represent small subgroups (~13% combined), the study may overemphasize atypical trajectories rather than those most representative of the population. Could a similar analysis be done on class 3 with stratification based on the level of EBW lost over the five years. This would greatly increase the power of the analyses. Alternately, the EBW could be expressed as a % and then the entire cohort could be stratified and analysed. It appears the initial latent class analysis has dictated and restricted the subsequent analyses.
- 5) The “reproducibility of the 13 feature score is based on either a 10 feature model or an 8 feature model. This does not make any sense, why and how were the models changed, I think

- this nullifies any claims of reproducibility within the analysis. If there were limited datasets on the later sample sets then the models should be rederived with the smaller number of features.
- 6) It is not clear what the predictors were in the broken-stick linear mixed-effects model fitted to the EBW data with knots at 6, 12, and 24 months, enabling interpolation across time points and estimating individual-specific weight trajectories over 7 years. This should be clarified.
 - 7) The reproducibility cohorts are very small (n=35) with quite different characteristics including initial BMI. While the second has little to do with the initial cohort or question being asked.
 - 8) Overall, the reproducibility is not convincing and I think the original model may be seriously overfitted given the number of original features and samples. To minimise this the whole model development process (feature selection and modelling) would need to be done within a cross validated framework.
 - 9) The metabolomic analyses are not adequately reported. There should be a supp table that shows all analytes with their annotations (or not), their missingness and their CVs across the cohort analysis. The Table should also include their association with the groups and the p values. Since most of the reported 62 associated analytes are not significant after correction, it is relevant what other similar analytes might have similar associations (or not). In any event it is important to get some perspective on the quality of the metabolomics analysis

Other comments

- 10) Line 94: how were the 80 participants selected from Class 2? Why wasn't the total ~100 used instead
- 11) Fig 1 a, Were the EBW between Class 2 and Class 3 across years post-surgery significantly different?
- 12) Line 130; how precisely can you say the metabolites are associated with sustained weight loss? The associations to me are just associations with %EBW5. So, is there an alternative way for interpretation of your results?
- 13) At line 153, the reported OR for younger age (0.95) appears to indicate a protective effect against weight regain rather than being a predictor of increased weight regain. Could the authors please revisit this interpretation
- 14) line 155: is the model limited to sex, baseline EBW, and age group? Could the authors clarify whether other potential predictors such as race, preoperative weight loss, or behavioral and dietary factors (if available) were considered, given their likely relevance to weight regain
- 15) line 156: could the authors clarify how the AUC was calculated? For example, was it derived within a 5- or 10-fold cross-validation framework, or using another approach
- 16) The authors stated that "The MetRS remained a strong predictor of weight regain in both unadjusted and multivariable-adjusted models (OR 2.63, 95% CI: 1.60, 4.31, p = 0.000132; Fig. 3 d), supporting its potential as a molecular marker for long-term surgical response". Does this mean the OR in Model 2, 3 and 4 is the same (which is unlikely) may be closer. Check the figure and clarify the statement in the manuscript
- 17) Line 175: "These findings suggest that MetRS captures a stable metabolic signature that persists after surgery". I think this is overinterpretation as metabolite levels are not measured at follow up and monitored for persistence. Consider re-wording
- 18) Line 316: is the sustained weight loss group in Post-RYGB Clinical Dataset defined in same way as in discovery dataset? And matched for same baseline characteristics?
- 19) Although the case cohort seems appropriate design, the finding could have benefited from including unoperated participants as controls.
- 20) Did the authors have access to dietary information? Was there any monitoring of diet and exercise levels in the two groups under study

Minor

- 1) The structure of the manuscript is somewhat difficult to follow, as it is hard to clearly differentiate between the Introduction, Results, and Discussion sections. Adding appropriate subheadings could help improve readability and guide the reader through the manuscript.
- 2) Line 39: “long-term weight outcomes” seems to me very broad. Specify the type of long term outcome your MetRS predicts
- 3) Line 111 I think you might need to clarify how redundancy was defined and whether compound identities were confirmed with authentic standards or based solely on MS/MS annotation
- 4) Fig 1 f, define red broken line. Also is y axis % EBW as mentioned in the text (line 119?
- 5) Line 133 uses specific procedure used instead of “MBS”
- 6) Figure 2 can be split into a table (a) and a figure (b and c panels)
- 7) Figure 2 metabolite names are missing on x axis and hard to read what is correlated with what
- 8) Figure 4. Label the top hit variance on the Manhattan plots
- 9) There is an extensive use of “—” throughout the document. some can be replaced by a space
- 10) Line 371: where are the results from LASSO?
- 11) Fig 4 d-g are not cited in manuscript
- 12) Line 103 and 109 change “Figure” to “Fig.” for consistency
- 13) Remove the figure borders

Reviewer #3 (Remarks to the Author):

I co-reviewed this manuscript with one of the reviewers who provided the listed reports. This is part of the Nature Communications initiative to facilitate training in peer review and to provide appropriate recognition for Early Career Researchers who co-review manuscripts.

Reviewer #4 (Remarks to the Author):

The ability to predict success of bariatric surgery would be helpful for individualization of therapeutic decisions re/ obesity management. Gu et al report results of analysis of the fasting plasma metabolome in participants within the LABS-2 cohort and its relation to weight change up to 5 years after RYGB. The authors identify a 13-metabolite risk score predictive of weight gain, and use this in an additional post-RYGB cohort and in a dataset incorporating extremes of BMI. Finally the identify significant genetic loci linked to individual metabolites. The authors acknowledge that it remains unclear whether identified metabolites are mechanistically related to weight change or consequences of altered weight trajectories.

For the primary analysis, the authors excluded class 3, which comprised 87% of the population. The prediction analysis then compared matched individuals from groups 1 and 2, but this further reduced sample size. This raises concerns about generalizability of results. It is unclear why a more continuous analysis of weight trajectory was not used to make greater use of the full cohort.

Expressing weight loss as a percentage of excess body weight is currently not a typical approach. Please provide explanation of the rationale for this approach.

For the 5 metabolites meeting the FDR<0.05 threshold, please describe the directionality of the

association. Why are these not included in the description of the 62 nominally associated metabolites?

Line 161 – why were only 10 of the 13 metabolites used to validate the score in the additional cohort?

Line 169 – why were only 8 components used in the validation for the Estonia Biobank?

Line 240 - The authors subsequently indicate that the 13-metabolite MRS demonstrated predictive utility in the 2 external cohorts, but this does not seem to be consistent with use of only 10 and 8 metabolites, respectively.

Please provide more context about BAIBA and its placement in amino acid metabolic pathways linked to obesity and metabolic disease based on prior studies. Can the authors show relationship between weight trajectories and nearby metabolites in the pathway, even if not significant. This analysis could potentially identify critical regulatory nodes.

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